

3 The curve C has equation

$$9y^2 - 3 \sinh^{-1}(xy) = 1 - 3 \ln 3.$$

(a) Show that, at the point $(4, \frac{1}{3})$ on C , $\frac{dy}{dx} = -\frac{1}{2}$. [4]

[illegible]



[5]

[illegible]